QSVM Track: Week 5 Progress Summary

Short form

2 August 2026

Scope. This is the Week 5 report for the QSVM track. Three of its results revise conclusions this track had previously drawn rather than merely extending them.

Data. Two malware datasets, each with a binary and a 15-class task. *CIC-MalMem-2022*: memory dumps, 55 forensic features; 58 596 balanced rows for the binary task, 29 298 malware-only rows over 15 families (1.7× imbalanced) for the 15-class one. *EMBER 2018*: static PE features, 2381 dimensions, on a 18 014-row extract of the published test set (88.9% malware); its 15-class task keeps the 15 largest families and downsamples each to the smallest, giving 14 325 rows at 955 per family.

Protocol. The quantum kernel costs $O(n^2)$ circuit evaluations, so every quantum result uses a stratified 1000-row subsample; that subsample and its five outer folds are persisted by the quantum runner and replayed by the classical one, so both are scored on identical rows and folds (800 train, 200 held out). Both consume one shared pipeline — variance filter, correlation filter at 0.95, standardisation, PCA to n_c components — fit on training-fold rows only. n_c abbreviates $n_{\text{components}}$ and fixes the classical feature count and the quantum qubit budget at once, which is what makes the comparison like-for-like.

Metric. Held-out macro-F1, meaned over five folds. Accuracy is not used as a headline: on EMBER binary at $n_c = 1$ the QSVM reaches 0.890 accuracy in three of five folds while the benign class's F1 is exactly 0.000 in those same folds, because 88.9% of rows are malware and the model labels everything malware.

1 What this week changes

Table 1: Items this track had left open, and their status now.

Position entering this week	**Now**	**Basis**
Classical comparison is SVM only	Resolved	Four models, $n_c \in \{1, 3, 6\}$, both tasks
CIC 15-class collapse unidentified	Attributed	Kernel concentration vs. an RBF control
No significance testing	Resolved	84 paired tests, all $p < 0.05$
Subsample fidelity assumed	Verified	KS rejections far below chance
SOREL blocked	Closed	Resume conditions recorded

A metric label to correct. Figures reported for the EMBER binary task up to now have been labelled "accuracy"; they are macro-F1. On that task the two differ by more than 0.4, and only one of them reflects whether the benign class is ever found.

2 Four-model classical baseline on EMBER

Every EMBER margin on record compared the QSVM against SVM alone. With random forest, XGBoost and LightGBM added, **the random forest beats SVM on the binary task at every** n_c — so those margins were measured against a sub-optimal baseline and are *understated*:

Table 2: EMBER 2018 held-out macro-F1. Best classical model in bold.

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
Binary						
$n_c = 1$	**0.5582**	0.5451	0.5093	0.5314	0.4532	0.4587
$n_c = 3$	**0.6575**	0.6465	0.6434	0.6306	0.4924	0.5084
$n_c = 6$	**0.6835**	0.6680	0.6593	0.6724	0.5138	0.5137
15-class						
$n_c = 1$	**0.5562**	0.5313	0.5410	0.5294	0.1688	0.1541
$n_c = 3$	0.7194	0.7089	0.7035	**0.7389**	0.3908	0.4797
$n_c = 6$	0.7903	0.7659	0.7709	**0.7930**	0.4232	0.5195

On binary, the quantum–classical gap grows from +0.0727 to +0.0995 ($n_c = 1$), +0.1222 to +0.1491 ($n_c = 3$) and +0.1587 to +0.1697 ($n_c = 6$). On the 15-class task SVM genuinely is the strongest baseline at $n_c \in \{3, 6\}$, so those two margins stand. Four of six cells move, all in the same direction: the margins recorded up to now flattered the quantum model.

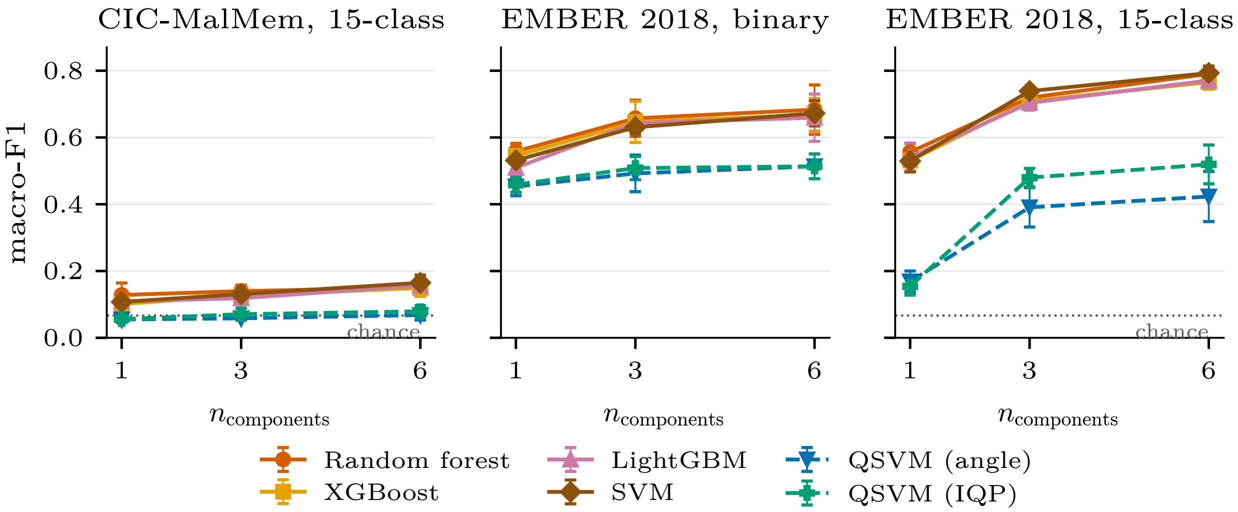

Figure 1: Macro-F1 against the shared feature/qubit budget; solid lines are classical, dashed the QSVM encodings. The gap on EMBER binary *widens* with n_c — the QSVM gains 0.055 from one to six components where the random forest gains 0.125. CIC binary is absent; see §4.

3 Why CIC 15-class collapses

It ruled out class imbalance and qubit count and closed with "the actual cause is unidentified". Two measurements separate a dataset effect from a kernel effect.

Feature geometry explains the dataset, not the kernel. On the identical projected features both frameworks consume, EMBER's families are $\approx 8\times$ more separable by Fisher ratio and up to $50\times$ on the worst-case family; CIC's silhouette is *negative at every* n_c, meaning the average CIC point lies closer to another family's centroid than its own. CIC's PCA nonetheless explains more variance (33–81% against 8–30%), so its variance sits in directions that do not discriminate family. This explains why *every* method does badly on CIC 15-class — and cannot explain the quantum-specific failure, since a classical SVM extracts 0.108–0.165 macro-F1 from these same features where the QSVM extracts 0.056–0.079.

Kernel geometry explains the kernel. Building a classical RBF Gram on the *identical* feature matrix separates "these features are hard" from "this kernel is bad".

- **Alignment separates the datasets, not the kernels.** On CIC the fidelity kernel lands 0.000069 *below* the constant-kernel floor — indistinguishable from an all-ones matrix — and the RBF control is below it too. On EMBER both clear it by ≈ 0.03.
- **Concentration separates the kernels, on CIC specifically.** RBF produces near-identical spread on both datasets (0.2212 vs. 0.2184), so CIC's features do not make a classical kernel concentrate. The fidelity kernel tracks RBF on EMBER ($0.92\times$) but collapses to **0.40**$\times$ its spread on CIC. Across the full sweep, EMBER's Gram is 1.9–2.6$\times$ more spread than CIC's at every setting.

Supported: the failure on CIC is a property of *the kernel*, not of the dataset. **Not supported:** that tuning `bandwidth` would repair it — that is the identified mechanism, never tuned on CIC beyond its default, and no sweep was run.

One tension, stated rather than resolved. Alignment reports that neither kernel's Gram carries label information on CIC, yet a classical SVM reaches 0.16 there. Alignment is a global unweighted average over all pairs; an SVM needs no such thing, only support vectors, a regularisation constant and class weights. It ranks dataset difficulty well and predicts achievable SVM performance poorly. The concentration result does not share this weakness, since it describes the Gram the SVM actually receives.

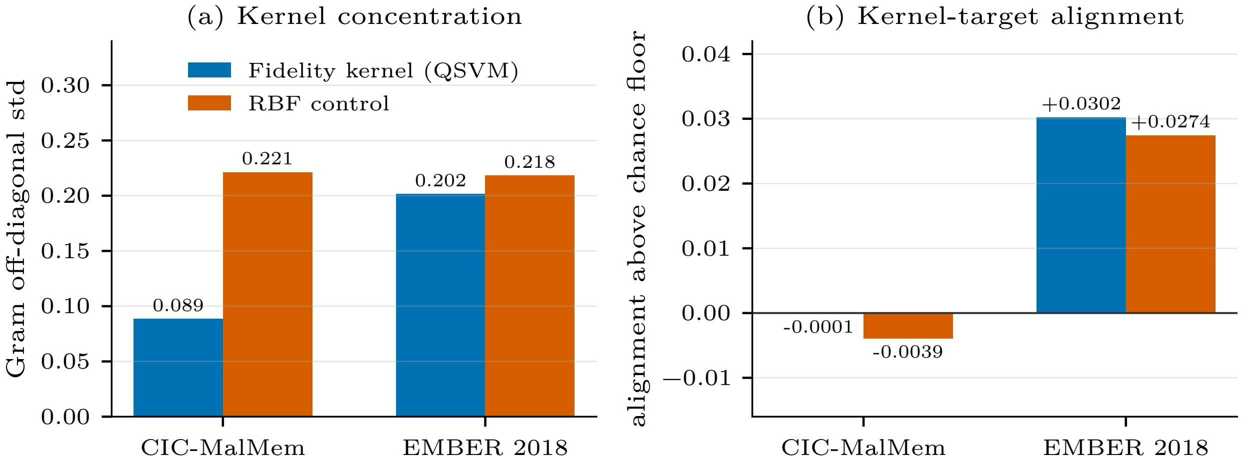

Figure 2: First outer training fold, $n_c = 6$, IQP encoding, 800 rows. (a) Off-diagonal Gram spread: as this collapses the Gram approaches a scaled identity and the SVM has nothing to separate on. (b) Kernel-target alignment relative to its floor — a constant kernel scores $1/\sqrt{15} \approx 0.258$ on 15 balanced classes, so only the excess carries meaning.

4 Paired significance tests

Specified in Week 1, built, unit-tested, and never run until now. Four weeks of "QSVM loses to classical" rested on mean differences with no test that any gap exceeded fold-to-fold noise.

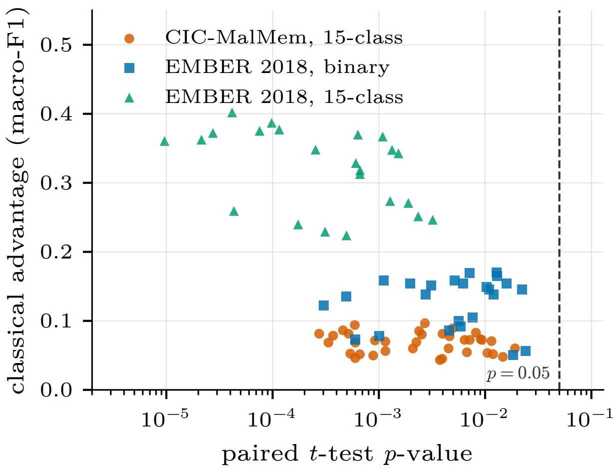

Figure 3: All 84 testable quantum–classical pairs. Every point lies left of $p = 0.05$ and above zero; the worst p-value across all 84 is 0.0241.

Wilcoxon returned 0.0625 **for all 84 pairs, and that is not a null result.** With five paired samples that is the *smallest attainable* two-sided p-value, so the test cannot reach 0.05 at this fold count regardless of effect size. All 84 hitting the floor means every pair achieved the most extreme rank configuration available. Six folds would put 0.03125 within reach; that, not a different test, is the remedy.

What could not be tested. *All CIC binary comparisons.* Every CIC binary run — classical *and* quantum — predates parent/child run tracking and carries no recoverable sweep boundary. The raw records show the QSVM at 0.92–1.00, consistent with the earlier "ties on binary" reading, but they resolve into two distinct sweeps per n_c with no identifier separating them; grouping them by time-gap clustering would invent a boundary the data does not record, and no attempt was made. The limitation is symmetric across frameworks. *McNemar between frameworks* needs per-sample predictions from both models; QSVM predictions were not persisted during these sweeps, and no sweep was re-run. Classical-versus-classical McNemar did run: on EMBER 15-class random forest beats XGBoost at $p = 0.0022$, while on CIC 15-class no pair separates at all.

Selecting comparable runs. Filtering on $(\text{dataset}, \text{task}, n_c, \text{model}, \text{encoding})$ fails twice over: the same configuration was legitimately swept several times across weeks (57 parameter groups have a size other than five), and for the QSVM *encoding is a per-fold outcome, not a configuration key* — the tuner selects it inside each fold, so one joint sweep can appear as four angle folds and one IQP fold. Runs are therefore grouped by parent run, keeping only sweeps that completed with exactly five folds; 123 of 125 nested sweeps are clean, and the two exceptions are excluded automatically rather than deleted.

5 Other closures

Subsample representativeness, specified in Week 2 and never executed, is now verified. Every quantum result uses a 1000-row stratified subsample forced by the $O(n^2)$ kernel cost. Read against its null expectation — a faithful subsample still produces $\approx \alpha \cdot n_{\text{features}}$ rejections by chance — CIC rejects *nothing* of 55 features where ≈ 2.75 were expected, and EMBER one of 2381 where ≈ 119 were. Maximum class-proportion drift is below 0.001 on both.

SOREL-20M is closed rather than deferred. Its labelling scheme was designed and validated over 19.7M rows and remains delivered work; the feature store was never fetched and no sweep was ever run — the two are independent. Neither disk (227 GiB free against 71.6 GiB needed) nor memory is the blocker; the store is a memory-mapped LMDB whose reads page through the OS cache. The blocker is the fixed 71.6 GiB transfer, unavoidable because LMDB offers no key-level remote access.

6 Summary

Classical models beat the QSVM on every testable configuration, now with $p < 0.05$ on all 84 rather than on mean differences alone, and the EMBER margins were understated rather than overstated. The one configuration where the quantum pipeline was ever competitive — CIC binary — cannot be significance-tested from what was logged, and we say so rather than dropping it quietly.

What is new is the mechanism. The fidelity kernel does not fail on CIC because CIC's features are hard: a classical RBF kernel on the identical features does not concentrate. It fails because the fidelity kernel collapses to $0.40\times$ the classical kernel's spread on that data — a statement about the method rather than the dataset, and the most transferable result this track has produced.